

# DHCP & NAT

Dynamic Host Configuration Protocol  
& Network Address Translation

Complete Study Notes | Networking Fundamentals

HTB Academy MODULE 2 PART 2

@manishburdak45

[https://www.linkedin.com/in/manish-burdak-718067380?utm\\_source=share\\_via&utm\\_content=profile&utm\\_medium=member\\_android](https://www.linkedin.com/in/manish-burdak-718067380?utm_source=share_via&utm_content=profile&utm_medium=member_android)

<https://github.com/manishburdak45>

# What is DHCP?

Dynamic Host Configuration Protocol (DHCP) is a network management protocol that automates the process of assigning IP addresses and other configuration parameters to devices on a network without manual intervention.

## **Automates IP Assignment**

No manual configuration needed for each device

## **Recycles Unused IPs**

When devices disconnect, IPs are returned to the pool

## **Prevents IP Conflicts**

Ensures every device gets a unique IP address

## **Configures More Than IP**

Also assigns subnet mask, gateway, DNS servers

# Why Do We Need DHCP?

## Without DHCP

- Admin must manually assign IP to every single device
- Easy to make mistakes — IP conflicts
- New device added? Configure it manually
- Huge workload in large networks
- IP wastage — unused IPs stay assigned

VS

## With DHCP

- IP assigned automatically when device connects
- Server ensures uniqueness — no conflicts
- Plug in a new device and it's ready instantly
- Scales effortlessly to thousands of devices
- Unused IPs are recycled back to the pool

# DHCP: Key Roles

## DHCP SERVER

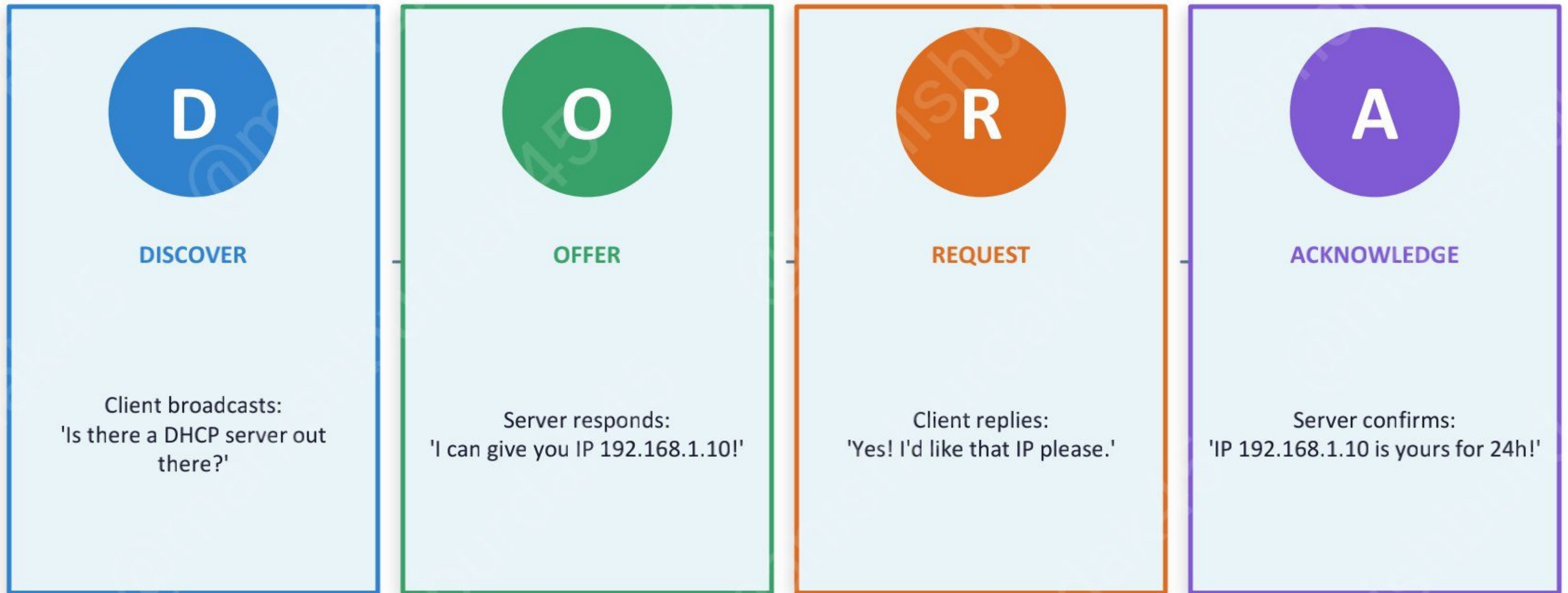
- A network device (router or dedicated server)
- Maintains a pool of available IP addresses
- Stores configuration parameters for clients
- Manages IP lease times and renewals
- Example: Your home Wi-Fi router

## DHCP CLIENT

- Any device that connects to the network
- Requests IP address and config from server
- Must renew IP before lease expires
- Can release IP when disconnecting
- Examples: Laptop, Phone, Smart TV, IoT device

# The DORA Process

How DHCP assigns an IP address in 4 steps



# DORA: Step-by-Step Details

## 1. DISCOVER

Client has no IP. Broadcasts to 255.255.255.255. Packet says: I need an IP address!

## 2. OFFER

Server picks an available IP from its pool and offers it, along with lease time, subnet mask, gateway, DNS.

## 3. REQUEST

Client broadcasts acceptance. Why broadcast? To notify other DHCP servers that it chose this offer.

## 4. ACKNOWLEDGE

Server finalizes the assignment. Client is now fully configured and can access the network.

# Real-Life Example: Alice's Laptop

Scenario: Alice joins a new office and connects her laptop to the network.

## 1 Laptop Connects

Alice's laptop has no IP. Sends a DHCP Discover broadcast: Anyone out there?

## 2 Server Responds

Office DHCP server receives the broadcast and offers IP 192.168.1.10 with 24hr lease.

## 3 Laptop Accepts

Alice's laptop sends DHCP Request: Yes, I'll take 192.168.1.10 please!

## 4 IP Confirmed

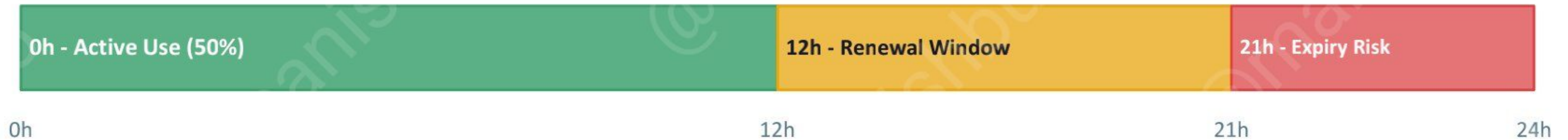
Server sends DHCP Acknowledge. Laptop has IP 192.168.1.10 and can browse the internet.

## 5 Lease Renewal

After ~12 hours, laptop auto-requests renewal. Server extends the lease for another 24 hours.

# IP Lease & Renewal

## IP Lease Timeline (Example: 24-hour lease)



### What is Lease Time?

The duration for which a DHCP server assigns an IP address. After expiry, the IP returns to the pool.

### Renewal Process

At 50% of lease time (e.g., 12h), client sends a unicast renewal request to the DHCP server.

### Rebinding

At 87.5% of lease time (e.g., ~21h), client broadcasts to any DHCP server if renewal failed.

### Lease Expiry

If no renewal, IP is released. Client must go through full DORA process again to get a new IP.

# DHCP Review Questions

**Q1: What protocol automates IP address configuration for devices on a network?**

Answer: DHCP - Dynamic Host Configuration Protocol

**Q2: What acronym describes the sequence of messages exchanged during the DHCP process?**

Answer: DORA - Discover, Offer, Request, Acknowledge

**Q3: What type of message does a client send to accept an IP address from a DHCP server?**

Answer: DHCP Request message

# Network Address Translation (NAT)

Solving the IPv4 Address Shortage

# What is NAT?

Network Address Translation (NAT) is a process carried out by a router that modifies source or destination IP addresses in packet headers as they pass through, translating private IPs to a single public IP.

## The IPv4 Problem

**~4.3B**

Total IPv4 Addresses

**8B+**

Devices on Earth

**1990s**

When shortage became critical

**NAT**

The rescue solution

Key Insight: NAT lets an entire home/office network share a single public IP address, just like one apartment block sharing one postal address!

# Public vs Private IP Addresses

## PUBLIC IP

- Globally unique - no two devices share
- Assigned by your ISP
- Accessible from anywhere on the internet
- Required to receive data from the internet

### Examples:

8.8.8.8 - Google DNS Server

142.251.46.174 - Google Web Server

## PRIVATE IP

- Used inside local networks only
- Not routable on the public internet
- Defined by RFC 1918
- Reused by millions of networks

### RFC 1918 Private Ranges:

10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

192.168.0.0 - 192.168.255.255

# How NAT Works - Step by Step



## NAT Translation Steps:

1. Laptop (192.168.1.10) sends request - Router changes source IP to 203.0.113.50
2. Router records mapping: 203.0.113.50:4444 <-> 192.168.1.10:5555 in NAT table

# Types of NAT

## Static NAT

One-to-one mapping of private IP to public IP. Each device gets a permanently assigned public IP.

Use case: Hosting a public web/mail server that needs a fixed, reachable IP.

*e.g. 192.168.1.10 always maps to 203.0.113.10*

## Dynamic NAT

Assigns a public IP from a pool of available addresses as needed. No permanent mapping.

Use case: Companies with a pool of public IPs, but fewer than total internal devices.

*e.g. 192.168.1.10 gets 203.0.113.X from pool when it connects*

## PAT / NAT Overload

Multiple private IPs share ONE public IP, differentiated by unique port numbers. Most common type!

Use case: Home routers, small offices - everyone shares one public IP.

*e.g. 192.168.1.10:5555 and 192.168.1.11:6666 both use 203.0.113.50 (different ports)*

# NAT: Benefits & Trade-Offs

## Benefits

### + Conserves IPv4 Addresses

Multiple devices share one public IP, saving the limited address space.

### + Security Layer

Internal network structure is hidden from the outside world.

### + Flexible Internal Addressing

You can use any private IP scheme internally without ISP approval.

## Trade-Offs

### - Complex Server Hosting

Hosting a public server behind NAT requires extra config (port forwarding).

### - Breaks End-to-End Protocols

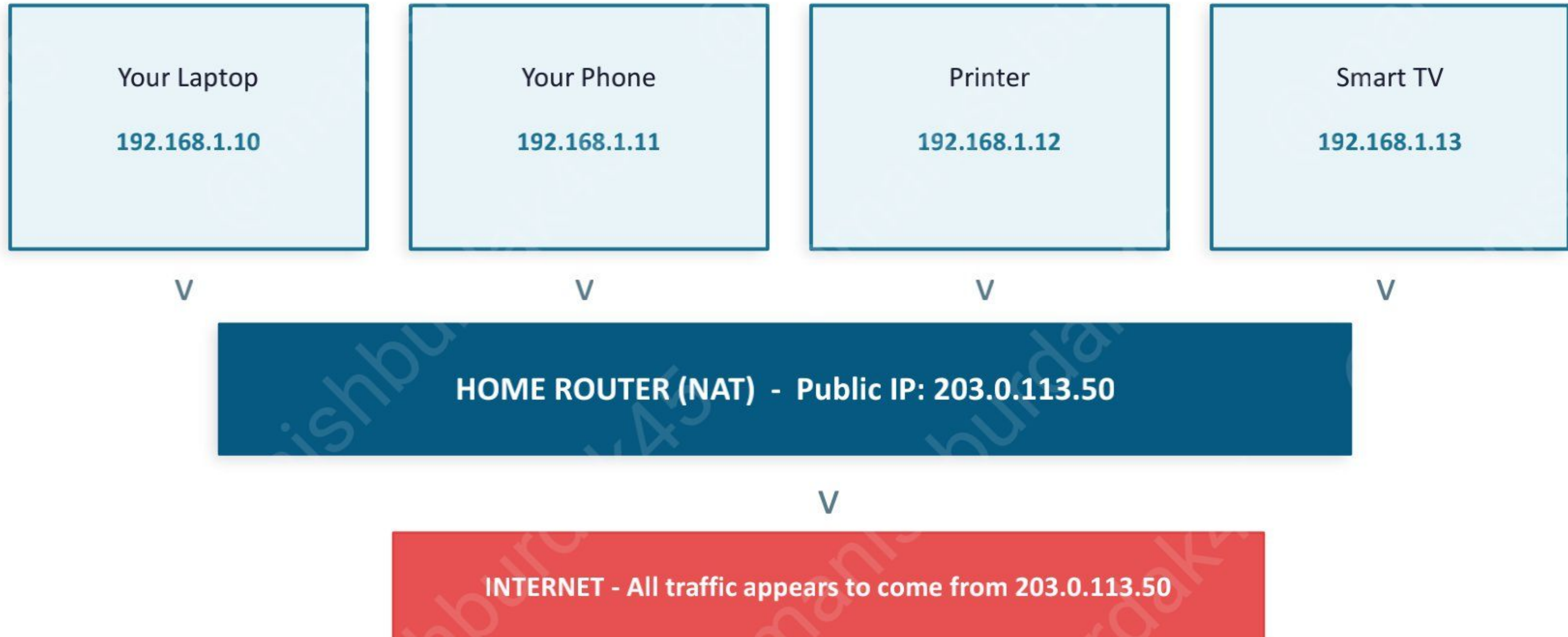
Some protocols like VoIP, P2P, or IPSec need special NAT handling.

### - Troubleshooting Complexity

Diagnosing connectivity issues is harder when IPs are being translated.

# Real-Life NAT: Your Home Network

Scenario: 4 devices at home sharing ONE public IP from your ISP



# NAT Review Questions

**Q1: What type of NAT allows multiple private IPs to share one public IP using unique port numbers?**

Answer: PAT (Port Address Translation) / NAT Overload

**Q2: What RFC specifies private IP ranges?**

Answer: RFC 1918

**Q3: Which NAT type involves a one-to-one mapping of private IP to public IP?**

Answer: Static NAT

**Q4: What type of NAT assigns a public IP from a pool as needed?**

Answer: Dynamic NAT

**Q5: What device typically performs NAT in a home network?**

Answer: The Router (home router/gateway)

# DHCP vs NAT - Quick Comparison

| Aspect       | DHCP                              | NAT                               |
|--------------|-----------------------------------|-----------------------------------|
| Purpose      | Assign IP addresses to devices    | Translate IPs for internet access |
| Who uses it  | Devices inside a local network    | Entire network (via router)       |
| Key RFC      | DHCP (RFC 2131)                   | NAT (RFC 1918 + RFC 3022)         |
| Process      | DORA: Discover-Offer-Request-Ack  | Modifies packet IP headers        |
| Where runs   | DHCP Server (router or dedicated) | Router / Gateway device           |
| Address type | Assigns private IP addresses      | Translates private and public IPs |
| Main benefit | Automates IP configuration        | Conserves public IP addresses     |

# Key Takeaways

## DHCP automates IP assignment

Uses the DORA process: Discover, Offer, Request, Acknowledge. Eliminates manual configuration.

## IPs are leased, not permanent

Lease has a time limit. Client must renew before expiry or go through DORA again.

## NAT solves the IPv4 shortage

Allows an entire network to share a single public IP by translating private IPs at the router.

## Private IPs stay private

RFC 1918 defines private ranges (10.x, 172.16.x, 192.168.x). Not routable on the internet.

## PAT is the most common NAT

Most home routers use PAT (NAT Overload) - multiple devices, one public IP, different port numbers.

# You're Ready!

DHCP & NAT - Study Complete